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CS 5830 Project 5: Naive Bayes Classifier

Presentation: CS5830 Project 5 Presentation Emberly and Brigham
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Introduction

Our project uses article summaries to predict article categories. This model could be used, for example, in editing or publishing software to automatically assign categories as authors make changes to article drafts. This model could also be extended to a newsfeed recommendation algorithm that proposes recommendations using categories as semantic signals.

Dataset

We scraped 158 publicly available articles by crawling category feeds (ex. https://www.usu.edu/today/section/arts-humanities/) on the Utah State Today website to extract plain-text article summaries. The data include the following categories: Arts & Humanities, Athletics, Business & Society, Campus Life, Health & Wellness, Land & Environment, Science & Technology, Teaching & Learning, and University Affairs. Each category is sufficiently represented in the data (the least-represented categories each have 17 articles). We preserved original categorizations in the dataset, although we did combine the Campus Life and University Affairs categories at the model layer because we found them to have no meaningful difference after a manual review.

Analysis Technique

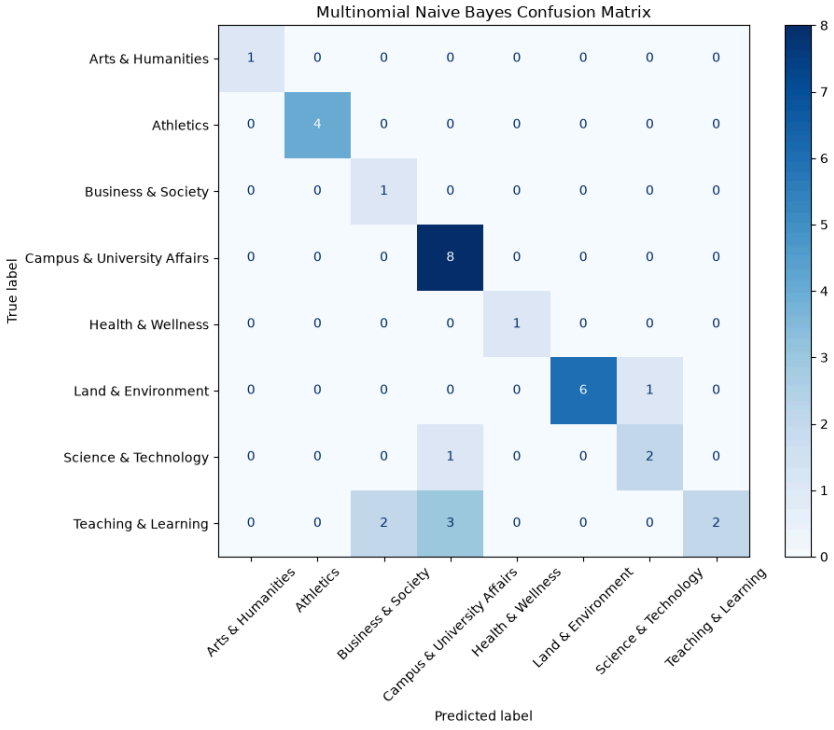
We trained a Multinomial Naive Bayes classifier using word counts for frequent, meaningful words as our feature set. We excluded English stop words and words that appear in at least 80% of articles as unmeaningful words. The Naive Bayes classifier assumes category-conditioned independence between each feature (in our case, word count), which is clearly violated as certain word pairs appear more often than others. However, this assumption provides a reasonable approximation for modeling the category of an article without introducing too much complexity. We chose Multinomial Naive Bayes instead of Gaussian Naive Bayes because the former is well-suited to features that involve discrete counts. The occurrence of a particular word can be thought of as a Bernoulli trial. We trained and tested the model using 80:20 train and test set partitioning.

Results

The predictive model that we created using a Multinomial Naive Bayes classifier was very precise. We calculated a precision score of .865, which shows that 86.5% of the model's positive predictions were correct. This means that the model has a relatively low rate of false positives. For example, if the article is not about a category such as athletics, the model is unlikely to

predict it as being about athletics. The recall score was lower than the precision score at .781, which means that the model successfully identified 78.1% of all articles belonging to each category. The precision and recall together give us an F-score of .765. On average, across all the different news categories, the model is very precise when assigning a label, but it misses about 21.9% of the articles that should have received that label. We chose to display the model's predictions vs true labels as a confusion matrix (see figure below).

Our predictive model is useful for many reasons. Labeling data can be time consuming and error prone. The university could use our model to label any future articles, or any old articles that don't include labels. This model could also be used for research. If someone wanted to find out how similar the Utah State Today newsfeed was to other Universities' newsfeeds, they could use our model to label other articles the same way as the Utah State articles are labeled. For our model we also collected all the most common words for each category, which is really interesting data that could be used for research purposes to find similar ideas or patterns across different news categories.



Technical

Overall, we didn't need to do a lot of formatting or cleaning for the dataset preparation. We extracted the plain-text article titles, descriptions, and categories from the Utah State Today website. We combined the Campus Life and University Affairs categories because we found that they had no meaningful difference. We chose to use a multinomial Bayes classifier to do our analysis because our dataset was working with words and word counts which are discrete features. A Multinomial Bayes Classifier is a good tool for text classification, which is why we

used it rather than Gaussian Naive Bayes, for sorting the USU news articles into their different categories.

We had several failed attempts and adjustments throughout the analysis process. At first, we didn't filter out common words and we saw really poor performance. We thought maybe we implemented the multinomial Bayes classifier incorrectly because of how poor it was, but after careful review we realized that the classifying technique doesn't work very well if you have common words included. We blacklisted English stop words, and words that appeared in 80% of the articles. We also got more sample articles by scraping the website directly instead of using the RSS feed, which had an article cap. We also found that some of the categories were indistinguishable from each other, so by combining those, we were able to improve our model's performance.